CATH PLAN · Case 4

62F, 70 kg · scheduled 12:30 · diagnostic cath ± PCI (exertional angina) · pathway: indeterminate, pre-cath plan

Synthetic teaching content. Not clinical advice. Not a medical device. DRAFT for the chair to rule on.

PRE-CATH PLAN: angiography has not been done. The band is a label from a proposed points rule, not a probability, and the packet lists equipment for both branches.

Thresholds: PROPOSED values, not yet verified by the chair. Template logic (likelihood weights and bands, prep rules, Imaging triggers, Heart Team factors, stage assignment, Labs' eGFR cutoff) is also unreviewed.

3	n/a	higher	0
flags from 4 agents	contrast ratio (Laskey 2007 discriminator 3.7, not a limit)	points band (8 pts, proposed rule)	devices out of stock

Contrast ratio = planned contrast (mL) ÷ eGFR. The published ratios use creatinine clearance, not eGFR (docs/EVIDENCE.md E1, E2): risk was flat below 2 and rose above 3 (Gurm 2011) and 3.7 was a discriminating cutoff (Laskey 2007). The plan shows both the eGFR ratio and, when age, sex, weight and creatinine are in the packet, a Cockcroft-Gault ratio, and applies the cutoffs to the higher. No guideline sets a limit; the chair confirms the basis.

Pre-cath findings (CCTA, stress, symptoms)

CCTA lesion	Vessel	Stenosis	Calcium
L1	distal RCA	60%	calcium moderate

Stress test: positive MPI, inferior wall · EF: 35% · Symptoms: typical angina, exertional dyspnea

Points driver (proposed points rule)	Points
CCTA 60%, intermediate stenosis (L1 distal RCA; E6)	+1
stress test positive (MPI, inferior wall)	+2
typical angina	+1
the positive stress result (inferior wall) is in the territory of the CCTA lesion	+2
low EF (35%)	+1
more than one symptom (typical angina, exertional dyspnea)	+1
Total · band: higher	**8**

A label from a points rule, not a probability. The chair decides whether to go on to PCI.

Lesion map (Imaging)

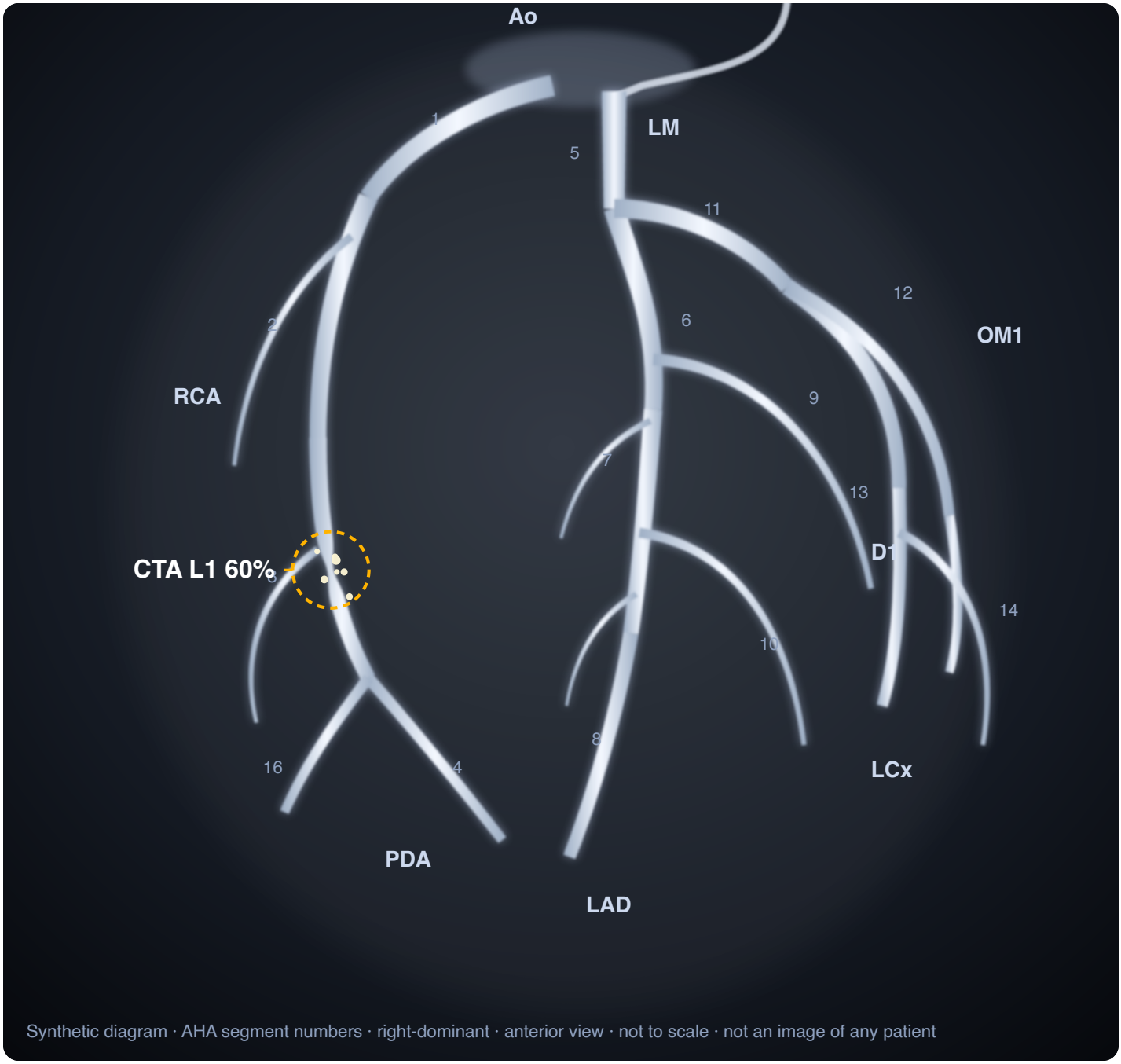
Lesion	Vessel	Stenosis	TIMI	Calcium	Flag
Angiography not yet performed: this is a pre-cath plan					

Angio findings: not assessed (angiography not yet performed). · Cath report: not in packet

Rules that fired

- Points band (proposed rule, 8 points): higher. The band word only labels a points range; it is not a probability and not a clinical pre-test probability. Chair to decide; the packet lists equipment for both branches below.

Pre-cath CCTA findings (synthetic, not yet confirmed by angiography)



L1 · distal RCA (AHA segment 3)
CCTA estimate 60% · calcium moderate · not yet confirmed by angiography

Drawn from the angio lines in the packet by tools/angio.py: right-dominant tree, anterior view, AHA segment numbers, lumen narrowed to the stated percentage, calcium shown as bright specks, distal filling dimmed when TIMI flow is below 3. Solid red ring = stenosis of 70% or more; dashed amber ring = under 70%, or a CCTA estimate not yet confirmed. A Medina 1,1,1 flag is drawn as a narrowing of about half the proximal LAD and LCx. A teaching diagram, not an image of any patient and not to scale.

Equipment manifest

Branch A · stays diagnostic	
Item (from the packet)	**Stock**
5F diagnostic catheters	6 on hand

Branch B · proceeds to PCI	
Item (from the packet)	**Stock**
6F guide	3 on hand
standard PCI tray	3 on hand

Prep check (devices against findings)

Scope	The packet lists Branch A and Branch B (band higher, proposed rule)
Stock	all named devices in stock

On the shelf, by class (from the STOCK line)	
Guide catheters	7F guide 3, 6F guide 3
Diagnostic catheters	5F diagnostic catheters 6
Wires	support wire 3
Balloons	NC balloon 3.5 mm 3, NC balloon 3.0 mm 3
Stents (DES)	DES 4.0x24 mm 2, DES 3.5x28 mm 2, DES 3.0x18 mm 2
Intravascular imaging	IVUS catheter 0 (resupply ETA 13:00)
Support devices	Impella CP 2

Questions for the chair (the plan asks, it does not choose a device):

- 1 lesion on CCTA and 0 stents named in the packet: chair to confirm what to stage.
- Calcium moderate on L1 distal RCA (CCTA): the packet names no calcium-modification device or intravascular imaging catheter. Chair to decide whether any is needed.
- No wire is named in the packet's device lines: chair to confirm.
- No balloon is named for the PCI branch: chair to confirm.
- EF 35% in the packet. The 2015 SCAI/ACC consensus names high-risk PCI (multivessel, left main, last conduit) with severely reduced EF but gives no EF cutoff (E10); this question fires at EF 35% or lower, the author's heuristic. The packet names no support device; the stock list shows: Impella CP 2. Whether support is needed is the chair's decision.
- Calcium or left main on CCTA, and IVUS catheter is 0 on hand with resupply ETA 13:00, after the scheduled 12:30 start: chair to decide how to proceed.

Flags from the agents

Imaging
- CCTA 60% · L1 distal RCA
- stress positive · inferior wall
- EF 35%

Labs ✅ nothing flagged

Meds ✅ nothing flagged

Equipment ✅ nothing flagged

Relay: Imaging → Equipment, Labs → Meds.

Summary and approvals

Ruling recorded	none yet
Flags	Imaging 3 · Labs 0 · Meds 0 · Equipment 0
Rules that fired	Points band (proposed rule, 8 points): higher. The band word only labels a points range; it is not a probability and not a clinical pre-test probability. Chair to decide; the packet lists equipment for both branches below.

For the chair to decide / check (the agents only flag):

- The points rule is proposed: the chair verifies it.
- The packet lists equipment for both branches: confirm stock for the branch actually taken.
- Thresholds in thresholds.json are proposed: the chair verifies them before this plan is used.
- Template logic (likelihood weights and bands, prep rules, Imaging triggers, Heart Team factors, stage assignment, Labs' eGFR cutoff) is unreviewed.

Approvals

No approval decisions recorded yet.

For the chair's use: print and mark.

- ☐ Approve as drafted
- ☐ Chair changes the plan (record the change)
- ☐ Equipment issue noted

Status

This draft is not approved until the chair presses Allow once on the Switch card.

AWAITING CHAIR

Sources: packet (day.md) · briefs from Imaging, Labs, Meds, Equipment · thresholds.json · evidence table docs/EVIDENCE.md (E1 to E16, mostly secondary summaries, chair verifies). Generated 2026-11-04 09:00 by plan.py (template, no model).
Provenance: template 7b83bc09dd · thresholds bea3b78f4a · plan.py fa5860885c · packet id n/a. "ref:" on a flag names a published criterion; it is not an instruction, and the criteria are unverified until the chair has checked them against the sources (docs/EVIDENCE.md).